

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12385132
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Hayashi; Ibuki et al.

Substrate processing apparatus and substrate processing method

Abstract

A substrate processing apparatus includes: a processing chamber; a rotary table rotatably provided inside the processing chamber; a stage that is rotatable relative to the rotary table at a position spaced apart from a rotation center of the rotary table; a process gas supplier provided above the rotary table; a separation gas supplier to supply a separation gas to separation regions, each of the separation regions being adjacent to a processing region; and a gas exhauster including exhaust ports communicating with an inside of the processing chamber, wherein the exhaust ports are provided in the processing region between the separation regions and are provided above the rotary table.

Inventors: Hayashi; Ibuki (Iwate, JP), Takeuchi; Yasushi (Iwate, JP), Honma; Manabu (Iwate, JP), Taguchi; Junnosuke (Iwate, JP)

Applicant: Tokyo Electron Limited (Tokyo, JP)

Family ID: 87559276

Assignee: Tokyo Electron Limited (Tokyo, JP)

Appl. No.: 18/163949

Filed: February 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230257877 A1	Aug. 17, 2023

Foreign Application Priority Data

JP	2022-023083	Feb. 17, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: C23C16/44 (20060101); C23C16/458 (20060101); H01L21/67 (20060101)

U.S. Cl.:

CPC C23C16/4412 (20130101); C23C16/4584 (20130101); H01L21/67017 (20130101);

Field of Classification Search

CPC: C23C (16/4412); C23C (16/4584); C23C (16/45519); C23C (16/4585); C23C (16/45574); C23C (16/46); C23C (16/52); C23C (16/44); C23C (16/4409); C23C (16/54); C23C (16/45551); C23C (16/545); C23C (14/568); C23C (16/4401); H01L (21/67017); H01L (21/6719); H01L (21/67196); H01L (21/67201); H01L (21/67173); H01L (21/67184); H01L (21/6776); H01L (21/67161); H01L (21/67155); H01L (21/67069); H01L (21/67748); H01L (21/67751); H01L (21/67745); H01J (37/32834)

USPC: 118/719; 118/715; 156/345.31; 156/345.32; 156/345.33; 156/345.34

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8043432	12/2010	Dip	118/719	C23C 16/54
9567669	12/2016	Cho	N/A	H01L 21/68771
11473196	12/2021	Ohashi	N/A	C23C 16/45551
2010/0294199	12/2009	Tran	118/723R	H01L 21/68785
2011/0212625	12/2010	Toyoda	438/758	C23C 16/4585
2012/0145078	12/2011	Huang	239/548	C23C 16/4412
2015/0361553	12/2014	Murakawa	156/345.55	C23C 16/45561
2016/0160349	12/2015	Cho	438/22	H01L 21/68771
2016/0172183	12/2015	Kamada	438/792	C23C 16/345
2017/0218515	12/2016	Shin	N/A	C23C 16/4412
2017/0287677	12/2016	Kato	N/A	H01J 37/3244
2021/0130950	12/2020	Hane	N/A	C23C 16/45574
2021/0156024	12/2020	Roh	N/A	C23C 16/4412
2021/0217651	12/2020	Taguchi	N/A	H01L 21/68792
2021/0301397	12/2020	Ohashi	N/A	C23C 16/345

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2010-114391	12/2009	JP	N/A
2013-026460	12/2012	JP	N/A
2013-044419	12/2012	JP	N/A
2021-111758	12/2020	JP	N/A

Primary Examiner: Zervigon; Rudy

Attorney, Agent or Firm: IPUSA, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is based on and claims priority to Japanese Patent Application No. 2022-023083 filed on Feb. 17, 2022, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present disclosure relates to a substrate processing apparatus and a substrate processing method.

Description of the Related Art

(3) Japanese Laid-Open Patent Publication No. 2021-111758 discloses a substrate processing apparatus including a rotary table which is rotatable in a processing chamber and a plurality of stages that are provided on the rotary table and are rotatable relative to the rotary table. For example, the substrate processing apparatus may rotate the rotary table and the stage while supplying a plurality of process gases into the processing chamber to deposit a film on a substrate placed on each stage.

(4) The substrate processing apparatus disclosed in Japanese Laid-Open Patent Publication No. 2021-111758 includes a separation gas supplier for supplying the separation gas between supplying the process gases, and a gas exhauster for exhausting each of the process gases and the separation gas from the processing chamber. The separation gas suppresses mixing of a first process gas and a second process gas in the processing chamber by separating the first process gas and the second process gas.

SUMMARY OF THE INVENTION

(5) According to one aspect of the present disclosure, a substrate processing apparatus includes: a processing chamber into which a substrate is transferred; a rotary table rotatably provided inside the processing chamber; a stage that is rotatable relative to the rotary table at a position spaced apart from a rotation center of the rotary table, wherein the stage is configured to receive the substrate thereon; a process gas supplier that is provided above the rotary table and that is configured to supply a process gas to the substrate placed on the stage; a separation gas supplier that is configured to supply a separation gas to separation regions, each of the separation regions being adjacent to a processing region to which the process gas is supplied; and a gas exhauster that includes one or more exhaust ports communicating with an inside of the processing chamber and that are configured to exhaust a gas through the one or more exhaust ports, wherein the one or more exhaust ports are provided in the processing region between the separation regions and are provided above the rotary table.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a longitudinal cross-sectional view illustrating a configuration example of a deposition apparatus according to an embodiment;

(2) FIG. 2 is a plan view illustrating a configuration in a processing chamber of the deposition apparatus;

(3) FIG. 3 is a perspective view illustrating a configuration of a rotary table and a stage of the deposition apparatus of FIG. 1;

(4) FIG. 4 is a flowchart illustrating a processing flow of a substrate processing method; and

(5) FIGS. 5A and 5B are schematic views illustrating exhaust operations of gases by a gas exhauster.

DESCRIPTION OF THE EMBODIMENTS

(6) Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. In each drawing, the same components are denoted by the same reference numerals, and overlapping descriptions are omitted.

(7) [Substrate Processing Apparatus]

(8) Referring to FIGS. 1 to 3, a deposition apparatus **1** for forming a film on a substrate **W**, which is an example of a substrate processing apparatus, will be described. FIG. 1 is a longitudinal cross-sectional view illustrating a configuration example of a deposition apparatus **1** according to an embodiment. FIG. 2 is a plan view illustrating a configuration in a processing chamber **11** of the deposition apparatus **1** illustrated in FIG. 1. In FIG. 2, a top plate is not illustrated for convenience of explanation. FIG. 3 is a perspective view illustrating a configuration of a rotary table **21** and a stage **211** of the deposition apparatus **1** in FIG. 1.

(9) The deposition apparatus **1** includes a processing unit **10**, a rotary drive device **20**, and a controller **90**.

(10) The processing unit **10** is configured to perform a deposition process of forming a film on the substrate **W**. The processing unit **10** includes a processing chamber **11**, a gas introduction section **12**, a gas exhauster **13**, a loading port **14**, a heating unit **15**, and a cooling unit **16**.

(11) The processing chamber **11** is a vacuum chamber capable of depressurizing its interior space. The processing chamber **11** is formed to be a flat housing having an approximately circular shape in a planar view, and can accommodate a plurality of substrates **W** in its interior space. The substrate **W** may be, for example, a semiconductor wafer. The processing chamber **11** includes a body **111**, a top plate **112**, a sidewall **113**, and a bottom plate **114** (FIG. 1). The body **111** has a cylindrical shape. The top plate **112** is detachably provided on the top of the body **111**. The body **111** and the top plate **112** are hermetically sealed by a seal **115**. The sidewall **113** has a cylindrical shape and is hermetically connected to the underside of the body **111**. The bottom plate **114** is hermetically connected to the bottom of the sidewall **113**.

(12) The gas introduction section **12** includes a raw material gas nozzle **121** and a reaction gas nozzle **122**, which are process gas suppliers **12a**, and includes separation gas nozzles **123** and **124**, which are separation gas suppliers **12b** (FIG. 2). The raw material gas nozzle **121**, the reaction gas nozzle **122**, and the separation gas nozzles **123** and **124** are spaced apart from each other along a circumferential direction of the processing chamber **11** (as indicated by arrow **A** in FIG. 2) above the rotary table **21**, which will be discussed below. In the illustrated example, the separation gas nozzle **123**, the raw material gas nozzle **121**, the separation gas nozzle **124**, and the reaction gas nozzle **122** are arranged in this order clockwise from the loading port **14** (rotation direction of the rotary table **21**). The raw material gas nozzle **121**, the reaction gas nozzle **122**, and the separation gas nozzles **123** and **124** each have gas inlet ports **121p**, **122p**, **123p**, and **124p** (FIG. 2) at their proximal ends for introducing various gases. The gas inlet ports **121p**, **122p**, **123p**, and **124p** are fixedly attached to the sidewall of the body **111** and protrude outwardly from the body **111**. The raw material gas nozzle **121**, the reaction gas nozzle **122**, and the separation gas nozzles **123** and **124** are inserted from the sidewall of the body **111** into the processing chamber **11** and extend radially inwardly of the body **111**. The raw material gas nozzle **121**, the reaction gas nozzle **122**, and the separation gas nozzles **123** and **124** are formed, for example, by quartz and arranged parallel to the rotary table **21**.

(13) The raw material gas nozzle **121** is connected to a source (not illustrated) of the raw material gas through a piping, a flow controller, and the like (not illustrated). Examples of the raw material gas include a gas containing silicon and a gas containing metal. The raw material gas nozzle **121** is provided with a plurality of discharge holes (not illustrated) that open toward the rotary table **21**. The plurality of discharge holes are arranged at intervals along the axial direction of the raw material gas nozzle **121**. The region below the raw material gas nozzle **121** is a raw material gas adsorption region **P1** in which the raw material gas is adsorbed on the substrate **W**.

(14) The reaction gas nozzle **122** is connected to a source (not illustrated) of the reaction gas

through a piping, a flow controller, and the like (not illustrated). Examples of the reaction gas include oxidized gas and nitride gas. The reaction gas nozzle **122** is provided with a plurality of discharge holes (not illustrated) that open toward the rotary table **21**. The plurality of discharge holes are arranged at intervals along the axial direction of the reaction gas nozzle **122**. The region below the reaction gas nozzle **122** is a reaction gas supply region **P2** in which the raw material gas adsorbed on the substrate **W** in the raw material gas adsorption region **P1** is oxidized or nitrided. In the present embodiment, the process gas for processing the substrate **W** is the raw material gas and the reaction gas.

(15) The separation gas nozzles **123** and **124** are connected to a source (not illustrated) of the separation gas through a piping, a flow controller, and the like (not illustrated). Examples of the separation gas include an inert gas such as argon (Ar) gas, nitrogen (N.sub.2) gas, and the like. The separation gas nozzles **123** and **124** are provided with a plurality of discharge holes (not illustrated) that open toward the rotary table **21**. The plurality of discharge holes are arranged at intervals along the axial direction of the separation gas nozzles **123** and **124**.

(16) As illustrated in FIG. 2, two projecting portions **17** are provided in the processing chamber **11**. The projecting portions **17** together with the separation gas nozzles **123** and **124** constitute the separation region **D** and are attached to the back surface of the top plate **112** so as to protrude toward the rotary table **21**. Each of the projecting portions **17** has a fan shape in a planar view with the apex of the fan shape being cut in an arc shape, with an inner arc connected to a protrusion **18** and an outer arc positioned along the sidewall of the processing chamber **11**.

(17) The gas exhauster **13** discharges the process gas and the separation gas supplied into the processing chamber **11** to the exterior of the processing chamber **11**. The configuration of the gas exhauster **13** will be described in detail below.

(18) The loading port **14** is provided on the sidewall of the body **111** (FIG. 2). At the loading port **14**, the substrate **W** is transferred between the rotary table **21** in the processing chamber **11** and a transfer arm **14a** outside the processing chamber **11**. The loading port **14** is opened and closed by a gate valve (not illustrated).

(19) The heating unit **15** includes a stationary shaft **151**, a heater support **152**, and a heater **153** (FIG. 1). The stationary shaft **151** has a cylindrical shape centered on the center of the processing chamber **11**. The stationary shaft **151** is provided inside the rotating shaft **23** of the rotary drive device **20** described below and through the bottom plate **114** of the processing chamber **11**. The heater support **152** is fixedly connected to the top of the stationary shaft **151** and has a disc shape. The heater **153** is supported on the upper surface of the heater support **152**. In addition to the upper surface of the heater support **152**, the heater **153** may be provided on the body **111**. The heater **153** generates heat by being supplied with power from a power source (not illustrated) to heat the substrate **W**.

(20) The cooling unit **16** includes fluid flow passages **161a** to **164a**, chiller units **161b** to **164b**, inlet lines **161c** to **164c**, and outlet lines **161d** to **164d**. The fluid flow passages **161a** to **164a** are formed within the body **111**, the top plate **112**, the bottom plate **114**, and the heater support **152**, respectively. The chiller units **161b** to **164b** output a temperature control fluid. The temperature control fluid output from the chiller units **161b** to **164b** flows through the inlet lines **161c** to **164c**, the fluid flow passages **161a** to **164a**, and the outlet lines **161d** to **164d** in this order and circulates. Accordingly, the temperatures of the body **111**, the top plate **112**, the bottom plate **114**, and the heater support **152** are adjusted. Examples of the temperature control fluid include water, a fluorine-based fluid such as Galden (registered trademark) and the like.

(21) The rotary drive device **20** includes a rotary table **21**, a housing box **22**, a rotating shaft **23**, a revolution motor **24**, and an outer barrel **25**.

(22) The rotary table **21** is provided in the processing chamber **11** and has a rotation center at the center of the processing chamber **11**. The rotary table **21** has a disc shape, for example, and is formed of quartz. A plurality of (for example, five) stages **211** are provided on the upper surface of

the rotary table **21** along the rotation direction (the circumferential direction). The rotary table **21** is connected to the housing box **22** via a connection **214** (FIG. 3).

(23) Each of the stages **211** has a disc shape slightly larger than the substrate **W** and is formed, for example, by quartz. The substrate **W** is placed on each of the stages **211**. Each of the stages **211** is connected to a rotation motor **213** via the rotating shaft **212** and is configured to be relatively rotatable with respect to the rotary table **21** (FIG. 1).

(24) The rotating shaft **212** connects the lower surface of the stage **211** to the rotation motor **213** contained within the housing box **22** and transmits power of the rotation motor **213** to the stage **211**. The rotating shaft **212** passes through a ceiling **222** of the housing box **22** and through the rotary table **21**. A seal **263** is provided near the penetration of the ceiling **222** of the housing box **22** to maintain airtightness within the housing box **22**. Examples of the seal **263** include a magnetic fluid seal.

(25) The rotation motor **213** is a rotation source that rotates the stage **211** via the rotating shaft **212** to rotate the substrate **W** around the center of the stage **211**. The rotation motor **213** is preferably a servomotor, for example.

(26) The connection **214** connects the lower surface of the rotary table **21** and the upper surface of the housing box **22** (FIG. 3). A plurality of connections **214** are provided along the circumferential direction of the rotary table **21**.

(27) The housing box **22** is provided below the rotary table **21** in the processing chamber **11**. The housing box **22** is connected to the rotary table **21** via the connection **214** and rotates integrally with the rotary table **21**. The housing box **22** may be configured to be liftable within the processing chamber **11** by means of a lifting mechanism (not illustrated). The housing box **22** includes a body **221** and the ceiling **222**.

(28) The body **221**, which has an upwardly open recess (FIG. 1), extends in the direction of rotation of the rotary table **21** to form an annular shape. The ceiling **222** is provided on the top of the body **221** to cover the opening of the recess in the body **221**. The body **221** and the ceiling **222** together form a rotation housing **223** that provides a space isolated from the inner space of the processing chamber **11**.

(29) The rotation housing **223**, which has a rectangular shape in a vertical cross-sectional view, extends in the direction of rotation of the rotary table **21** to form an annular shape. The rotation housing **223** accommodates the rotation motor **213**. A communication passage **224** is formed in the body **221** for communicating the rotation housing **223** with the outside of the deposition apparatus **1**. Accordingly, outside air is introduced into the rotation housing **223** from the outside of the deposition apparatus **1**, which cools the inside of the rotation housing **223** and also maintains an atmospheric pressure therein. In order to rotatably arrange the rotation housing **223**, the processing chamber **11** includes a rotation source housing space **19** surrounded by the sidewall **113**, the bottom plate **114**, and the heating unit **15**.

(30) The rotating shaft **23** is fixedly connected to a lower part of the housing box **22**. The rotating shaft **23** is provided through the bottom plate **114** of the processing chamber **11**. The rotating shaft **23** transmits the power of the revolution motor **24** to the rotary table **21** and to the housing box **22**, to rotate the rotary table **21** and the housing box **22** together. A seal **154** is provided between the outer wall of the stationary shaft **151** and the inner wall of the rotating shaft **23** of the rotary drive device **20**. Thus, the rotating shaft **23** rotates with respect to the stationary shaft **151** while maintaining the airtightness inside the processing chamber **11**. Examples of the seal **154** include a magnetic fluid seal.

(31) The outer barrel **25** of the rotary drive device **20** is connected to a central lower surface of the bottom plate **114** of the processing chamber **11**. The outer barrel **25**, together with the stationary shaft **151** of the processing chamber **11**, supports the processing chamber **11**. A seal **116** is provided between the rotating shaft **23** and the outer barrel **25** to maintain the airtightness within the processing chamber **11**. Examples of the seal **116** include a magnetic fluid seal.

(32) The gas introduction section **12** includes an inert gas supplier **27** for supplying the inert gas to the shaft space between the rotating shaft **23** and the outer barrel **25**. The inert gas supplier **27** includes a gas supply **27a** outside the processing chamber **11** and supplies the inert gas into the shaft space under the control of the controller **90**. The shaft space communicates with the rotation source housing space **19**. The inert gas supplied to the shaft space flows into the rotation source housing space **19**.

(33) A passage **231** is formed inside the rotating shaft **23**. The passage **231** is connected to the communication passage **224** of the housing box **22** and functions as a fluid flow passage for introducing air into the housing box **22**. The passage **231** also functions as a wiring duct for introducing power lines and signal lines into the housing box **22**, for driving the rotation motor **213**. The passages **231** are provided, for example, to be same in number as the rotation motors **213**.

(34) The controller **90** controls each part of the deposition apparatus **1**. The controller **90** may be, for example, a computer having a processor, a memory, and an I/O interface, which are not illustrated. Computer programs for operating each part of the deposition apparatus **1** are stored in the memory. The memory may be, for example, a flexible disk, a compact disk, a hard disk, a flash memory, a DVD, and the like.

(35) [Configuration of Gas Exhauster **13**]

(36) Next, the configuration of the gas exhauster **13** of the deposition apparatus **1** will be described in detail with reference to FIGS. **1** and **2**. In the gas exhauster **13**, a plurality of exhaust ports **131a**, **131b**, and **137a** are provided above the substrate **W** placed on the stage **211** in the raw material gas adsorption region **P1** and in the reaction gas supply region **P2**, for evacuating gas from the interior space of the processing chamber **11**. Accordingly, in the raw material gas adsorption region **P1** and in the reaction gas supply region **P2**, the flow of gas to the lower surface side of the stage **211** is prevented. Meanwhile, the gas exhauster **13** allows the supplied separation gas to flow to the lower surface side of the stage **211** in the plurality of separation regions **D**.

(37) Specifically, the gas exhauster **13** includes an outer exhaust structure **131** which is circularly disposed along a circumferential direction of the processing chamber **11** radially outside of the rotary table **21**, and an axial exhauster **137** which is provided at an adjacent position of the protrusion **18** in the raw material gas adsorption region **P1**.

(38) The outer exhaust structure **131** includes a ring member **132** provided in contact with the bottom wall of the body **111**, a spacer **133** stacked on the ring member **132**, and a cover member **134** stacked on the spacer **133**.

(39) The ring member **132** has an endless shape that is circularly disposed along the inner circumferential surface of the body **111** in a plane view. The ring member **132** has an upwardly open recess in a side cross-sectional view, having a gas flow passage **132a** therein. The gas flow passage **132a** is continuous around the entire circumference of the ring member **132**.

(40) The ring member **132** also includes an external communication port **132b** for communicating with the gas flow passage **132a**, at a predetermined position on the bottom wall. The external communication port **132b** communicates with the flow passage of an exhaust pipe **135** provided outside the processing chamber **11** via a communication hole **111a** formed in the body **111**. The exhaust pipe **135** is provided with an exhaust device **136** that causes gas to flow in the exhaust pipe **135**. The exhaust device **136** may include, for example, a pump to suction the gas in the processing chamber **11**, a flow control valve to adjust the flow rate of the gas to be exhausted, and the like.

(41) The spacer **133** is provided around the entire circumference of the ring member **132** and covers the top of the gas flow passage **132a**. The spacer **133** is positioned exactly laterally (at the same height) of the rotary table **21** in a side cross-sectional view. The spacer **133** has its inner peripheral surface positioned radially outside of the inner peripheral surface of the ring member **132** and is non-contact with the rotary table **21**. Accordingly, by the spacer **133**, the cover member **134** is positioned slightly higher than the rotary table **21**. At the appropriate location on the spacer **133**, an exhaust passage **133a** is formed for communicating the first exhaust port **131a** and the

second exhaust port **131b** of the cover member **134**, and the gas flow passage **132a** of the ring member **132**, which will be described later.

(42) The cover member **134** is formed into an annular plate which is provided around the entire circumference of the spacer **133**. In the assembled state of the outer exhaust structure **131**, the lower surface of the cover member **134** is positioned above the upper surface of the rotary table **21** in a vertical direction. The cover member **134** has an inner edge projecting radially inwardly relative to the inner peripheral surface of the spacer **133**. Thus, the inside of the cover member **134** is positioned non-contact above the outer edge of the rotary table **21**.

(43) The outer exhaust structure **131** includes the first exhaust port **131a** and the second exhaust port **131b** passing through the cover member **134** (see also FIG. 2). Because the cover member **134** is at a higher position than the rotary table **21**, the first exhaust port **131a** and the second exhaust port **131b** can exhaust gases in the processing chamber **11** above the rotary table **21** in the vertical direction. The gas taken into the first exhaust port **131a** and the second exhaust port **131b** flows into the gas flow passage **132a** via the exhaust passage **133a** and is further discharged into the flow passage of the exhaust pipe **135** via the external communication port **132b** and the communication hole **111a**.

(44) As illustrated in FIG. 2, the first exhaust port **131a** is formed adjacent to and radially outside of the raw material gas adsorption region **P1**, and is formed in the first exhaust region **E1**, which is in communication with the raw material gas adsorption region **P1**. The first exhaust port **131a** is positioned at the downstream side of the rotation direction (clockwise in FIG. 2) of the rotary table **21** in the raw material gas adsorption region **P1**. Thus, the first exhaust port **131a** passes the raw material gas supplied to the raw material gas adsorption region **P1** and the separation gas supplied to the separation region **D** adjacent to the raw material gas adsorption region **P1**.

(45) The second exhaust port **131b** is formed adjacent to and radially outside of the reaction gas supply region **P2**, and is formed in the second exhaust zone **E2**, which is in communication with the reaction gas supply region **P2**. The second exhaust port **131b** is positioned at the downstream side of the rotation direction (clockwise in FIG. 2) of the rotary table **21** in the reaction gas supply region **P2**. Thus, the second exhaust port **131b** passes the reaction gas supplied to the reaction gas supply region **P2** and the separation gas supplied to the reaction gas supply region **P2**.

(46) As illustrated in FIGS. 1 and 2, the axial exhauster **137** is provided adjacent to the radially inward part of the raw material gas adsorption region **P1**. The axial exhauster **137** includes an upper exhaust port **137a** formed in the top plate **112** and an exhaust port **138** communicating with the upper exhaust port **137a**. The axial exhauster **137** may be provided not only in the raw material gas adsorption region **P1**, but also in the radially inward part of the reaction gas supply region **P2**, or only in the reaction gas supply region **P2**.

(47) The upper exhaust port **137a** is positioned radially inward relative to the stage **211** of the rotary table **21** and penetrates the top plate **112**. Accordingly, the upper exhaust port **137a** does not face the substrate **W** placed on the stage **211**. The upper exhaust port **137a** is formed at a position overlapping the imaginary radial line extending from the center of the processing chamber **11** (the rotation center of the rotary table **21**) toward the first exhaust port **131a**.

(48) The exhaust port **138** protrudes upward from the top plate **112** to connect with the exhaust line **139** outside the processing chamber **11**. The exhaust port **138** includes an internal flow passage **138a** that communicates the upper exhaust port **137a** and the flow passage of the exhaust line **139**. The exhaust line **139** is connected to the exhaust device **136** outside the processing chamber **11**.

(49) The gas exhauster **13** described above can guide the raw material gas and the separation gas above the rotary table **21** in the raw material gas adsorption region **P1**. Similarly, the gas exhauster **13** can guide the reaction gas and the separation gas above the rotary table **21** in the reaction gas supply region **P2**.

(50) [Substrate Processing Method]

(51) The deposition apparatus **1** according to the present embodiment is basically configured as

described above, and the operation (a substrate processing method) will be described below with reference to FIGS. 4 and 5. FIG. 4 is a flowchart illustrating a processing flow of the substrate processing method. FIGS. 5A and 5B are schematic views illustrating exhaust operations of gases by the gas exhauster 13, wherein FIG. 5A illustrates the flow of the gas in the raw material gas adsorption region P1 and FIG. 5B illustrates the flow of the gas in the separation region D.

(52) As illustrated in FIG. 4, the substrate processing method includes a loading step (step S1), a depressurization step (step S2), a rotation step (step S3), a separation gas supply step (step S4), a process gas supply step (step S5), and an unloading step (step S6). The controller 90 controls each configuration of the deposition apparatus 1 and performs each step at appropriate timing.

(53) In the loading step (step S1), under the control of the controller 90, the transfer arm 14a enters through the loading port 14 with the gate valve open, and the substrate is transferred (placed) on each of the stages 211 of the rotary table 21 provided in the processing chamber 11. Under the control of the controller 90, after placing the substrate W on the stage 211, the gate valve is closed and the processing chamber 11 is sealed.

(54) In the depressurization process (step S2), under the control of the controller 90, the gas exhauster 13 is operated to exhaust the gas inside the processing chamber 11 through each of the exhaust ports (the first exhaust port 131a, the second exhaust port 131b, and the upper exhaust port 137a), which communicate with the inside of the processing chamber 11. Accordingly, the pressure of the space inside the processing chamber 11 is reduced to a predetermined level. Then, under the control of the controller 90, by continuously controlling the operation of the gas exhauster 13, the reduced pressure in the depressurization step is maintained until the start of the unloading step.

(55) In the rotation step (step S3), under the control of the controller 90, the rotary table 21 is revolved. Under the control of the controller 90, each of the stages 211 on which the substrate W is placed is rotated relative to the rotary table 21.

(56) In the separation gas supply step (step S4), under the control of the controller 90, the separation gas is supplied from each of the separation gas nozzles 123 and 124 provided within the processing chamber 11 to form a plurality of separation regions D.

(57) Under the control of the controller 90, as a deposition process (a substrate processing) for the substrate W, the process gas supply step (step S5) is performed in which the raw material gas is supplied from the raw material gas nozzle 121 and the reaction gas is supplied from the reaction gas nozzle 122. The process gas supply step may be performed simultaneously with the separation gas supply step.

(58) In each substrate W, which revolves and rotates by the rotary table 21, the raw material gas is adsorbed when it passes through the raw material gas adsorption region P1, and the reaction gas reacts with the adsorbed raw material gas when it passes through the reaction gas supply region P2. In the deposition process, by continuing steps S2 to S5 for a predetermined period of time, a film having a desired thickness is formed on the surface of the substrate W.

(59) When the deposition process is completed, under the control of the controller 90, the depressurization of the processing chamber 11 and the supply of the process gas are stopped, the pressure inside the processing chamber 11 is increased, and the supply of the separation gas is stopped in an appropriate pressure condition. Under the control of the controller 90, the rotation of the rotary table 21 and each of the stages 211 are stopped so that the substrate W on each of the stages 211 can be removed.

(60) In the unloading step (step S6), under the control of the controller 90, the transfer arm 14a enters through the loading port 14 with the gate valve open, and the substrate W is unloaded from each of the stages 211 of the rotary table 21.

(61) In the depressurization step (step S2) to the process gas supply step (step S6) of the substrate processing method described above, the gas exhauster 13 continues to evacuate the gas in the processing chamber 11. In particular, in the process gas supply step, the process gas supplied into the processing chamber 11 is continuously evacuated through the plurality of exhaust ports 131a,

131b, and **137a**. Next, the flow of the process gas and the separation gas when the gases are exhausted by the gas exhauster **13** (the depressurization step) in the deposition process (when the process gas and the separation gas are supplied) will be described.

(62) As illustrated in FIG. 5B, in the substrate processing, the separation gas nozzles **123** and **124** discharge the separation gas from above the rotary table **21** toward the substrate W placed on the stage **211**. As described above, because the projecting portion **17** is provided in the separation region D, the space between the projecting portion **17** and the rotary table **21** is narrowed. This increases the internal pressure of the separation gas and prevents the inflow of the process gas (the raw material gas and the reaction gas) into the separation region D.

(63) The inert gas supplier **27** supplies the inert gas to the rotation source housing space **19** below the body **111**. The inert gas flows into the space between the rotary table **21** and the heater **153** (hereinafter referred to as an under-table space **28**) through the gap between the rotating shaft **212**, which axially supports the stage **211**, the body **111**, and the heating unit **15**.

(64) A portion of the separation gas supplied to the separation region D flows radially outwardly above the rotary table **21**. The other portion of the separation gas passes through the gap between the rotary table **21** and the stage **211**, goes around to the lower surface side of the stage **211**, and further flows to the under-table space **28**. In the under-table space **28**, the separation gas is mixed with the inert gas from the rotation source housing space **19**. The separation gas and the inert gas flow radially outward in the under-table space **28**.

(65) Meanwhile, as illustrated in FIG. 5A, in the raw material gas adsorption region P1, the raw material gas nozzle **121** discharges the raw material gas from above the rotary table **21** toward the substrate W placed on the stage **211**. Accordingly, the raw material gas is adsorbed to the upper surface of the substrate W. The inert gas flows into the under-table space **28** in the raw material gas adsorption region P1, from the rotation source housing space **19** below the body **111**.

(66) The gas exhauster **13** according to the present embodiment includes the first exhaust port **131a** at a radially outward adjacent position and the upper exhaust port **137a** at a radially inward adjacent position, in the raw material gas adsorption region P1. The first exhaust port **131a** and the upper exhaust port **137a** are arranged above the rotary table **21** in a vertical direction (proximate to the raw material gas nozzle **121**). Accordingly, in the raw material gas adsorption region P1, a flow is formed in which the gas is directed upward from the under-table space **28**.

(67) Specifically, a portion of the inert gas flowing into the under-table space **28** moves to the lower surface side of the stage **211**, and is directed to the space above the rotary table **21** through the space between the rotary table **21** and the stage **211**. The other portion of the inert gas flowing into the under-table space **28** moves radially outward of the under-table space **28**, and is directed to the space above the rotary table **21** through the space between the rotary table **21** and the ring member **132**.

(68) In the space above the rotary table **21**, the raw material gas supplied from the raw material gas nozzle is mixed with the inert gas. The raw material gas and the inert gas are directed to the first exhaust port **131a** and the upper exhaust port **137a** under suction action of the exhaust device **136** via the first exhaust port **131a** and the upper exhaust port **137a**. The first exhaust port **131a** is positioned radially outside relative to the rotary table **21** and at a higher position than the rotary table **21**, so that the raw material gas and the inert gas can be stably guided upward. Similarly, the upper exhaust port **137a** is positioned radially inward relative to each of the stages **211** and at a higher position than the rotary table **21**, so that the raw material gas and the inert gas can be stably guided upward.

(69) Here, in a case where the first exhaust port is positioned below the rotary table **21** in the vertical direction, the raw material gas supplied to the space above the rotary table **21** moves downward. At this time, the raw material gas is more likely to go around into the gap between the stage **211** and the rotary table **21**. In particular, in the configuration of rotating the stage **211**, there is inevitably a gap between the stage **211** and the rotary table **21**. Therefore, when the first exhaust

port is lower than the rotary table **21**, the raw material gas goes around to the lower surface side of the stage **211**. The raw material gas that goes around the lower surface side of the stage **211** attaches to the lower surface of the stage **211**, thereby increasing the possibility of affecting the rotation of the stage **211**.

(70) In contrast, in the deposition apparatus **1** according to the present embodiment, because the first exhaust port **131a** is provided above the rotary table **21** in the vertical direction, the raw material gas can be prevented from going around to the space between the stage **211** and the rotary table **21**. That is, the inert gas (or the separation gas from the separation region D) is directed into the gap between the stage **211** and the rotary table **21**, and these gases flow directly into the space above the rotary table **21** and are directed to the upper first exhaust port **131a**.

(71) In particular, in the deposition apparatus **1** according to the present embodiment, the upper exhaust port **137a** is provided radially inwardly, so that the raw material gas and the inert gas on the rotation center side of the rotary table **21** can be smoothly discharged from the upper exhaust port **137a** without directing the raw material gas and the inert gas radially outwardly. Accordingly, the raw material gas is not left on the rotation center side of the rotary table **21**, and thus the raw material gas can be prevented from going around to the lower surface side of the stage **211** more reliably.

(72) In addition, in the deposition apparatus **1**, the reaction gas and the inert gas can be guided upwardly by the second exhaust port **131b** provided above the rotary table **21** in the vertical direction, in the reaction gas supply region P2. Accordingly, in the reaction gas supply region P2 also, the raw material gas can be prevented from going around to the lower surface side of the stage **211**.

(73) The deposition apparatus **1** may be a configuration in which the first exhaust port **131a** of the raw material gas adsorption region P1 is higher than the rotary table **21**, while the second exhaust port **131b** of the reaction gas supply region P2 is lower than the rotary table **21**. Conversely, the deposition apparatus **1** may be a configuration in which the second exhaust port **131b** of the reaction gas supply region P2 is higher than the rotary table **21**, while the first exhaust port **131a** of the raw material gas adsorption region P1 is lower than the rotary table **21**. Even when one of the raw material gas adsorption region P1 and the reaction gas supply region P2 evacuates above the rotary table **21** and the other evacuates below the rotary table **21**, mixing of the raw material gas and the reaction gas on the lower surface side of the rotary table **21** can be prevented. Therefore, it is possible to prevent the deposition on the lower surface side of the rotary table **21**, and the generation of particles can be prevented.

(74) The deposition apparatus **1** may include a plasma processing region in which the substrate W is plasma-processed on the rotation path of the rotary table **21**. For example, the plasma processing region is provided between the reaction gas supply region P2 and the separation region D on the downstream side of the rotation direction (clockwise) of the rotary table **21**.

(75) The technical ideas and effects of the present disclosure described in the above embodiments will be described below.

(76) A substrate processing apparatus (a deposition apparatus **1**) according to a first aspect of the present disclosure includes: a processing chamber **11** for processing a substrate W transferred therein; a rotary table **21** rotatably provided inside the processing chamber **11**; a stage **211** that is rotatable relative to the rotary table **21** at a position spaced apart from a rotation center of the rotary table **21**, and that is configured to receive the substrate W thereon; a process gas supplier **12a** that is provided above the rotary table **21** and that is configured to supply a process gas to the substrate W placed on the stage **211**; a separation gas supplier **12b** that is configured to supply a separation gas to separation regions, each of the separation regions D being adjacent to a processing region to which the process gas is supplied; and a gas exhauster **13** that includes one or more exhaust ports **131a**, **131b**, and **137a** communicating with an inside of the processing chamber **11** and that are configured to exhaust a gas through the one or more exhaust ports **131a**, **131b**, and **137a**, wherein

the one or more exhaust ports **131a**, **131b**, and **137a** are provided in the processing region between the separation regions D and are provided above the rotary table **21**.

(77) According to the above, in the substrate processing apparatus (the deposition apparatus **1**), the exhaust ports **131a**, **131b**, and **137a** are provided above the rotary table **21**, so that the process gas supplied to the substrate W can be guided upward to the exhaust ports **131a**, **131b**, and **137a**. Accordingly, it is possible to prevent the process gas supplied into the processing chamber **11** from going around to the lower surface side of the stage **211**, thereby preventing the process gas from being attached on the lower surface of the stage **211**. In addition, the process gas which is prevented from going around to the stage **211** is guided toward the exhaust ports **131a**, **131b**, and **137a** by the separation region D of the separation gas in the space above the substrate W, so that the process gas is smoothly exhausted. For example, in the configuration of supplying a plurality of process gases, the mixing of the plurality of process gases can be more reliably prevented.

(78) At least one of the one or more exhaust ports **131a** and **131b** may be provided outside the outer edge of the rotary table **21**. Accordingly, the exhaust ports **131a** and **131b** can well guide the process gas toward the outer edge of the rotary table **21** in the centrifugal direction without blocking the supply of the process gas to the substrate W on the stage **211**.

(79) The gas exhauster **13** may include an outer exhaust structure **131** that is provided outside the outer edge of the rotary table **21** and that is circularly disposed along an inner peripheral surface of the processing chamber **11**. The outer exhaust structure **131** may include a member (the cover member **134**) provided above the rotary table **21**, the member having the one or more exhaust ports **131a** and **131b** outside the outer edge of the rotary table **21**. Accordingly, in the substrate processing apparatus (the deposition apparatus **1**), the exhaust ports **131a** and **131b** can be easily provided above the rotary table **21**.

(80) At least one of the one or more exhaust ports **137a** may be provided closer to the rotation center of the rotary table **21** than is the stage **211**. Accordingly, in the substrate processing apparatus (the deposition apparatus **1**), the process gas can be easily exhausted even on the rotation center side of the rotary table **21**.

(81) The processing chamber **11** may also include a body **111** having a recess, and a top plate **112** that closes an upper side of the body **111**. The at least one of the one or more exhaust ports **137a**, which is provided closer to the rotation center of the rotary table **21** than is the stage, may be provided on the top plate **112**. Accordingly, in the substrate processing apparatus (the deposition apparatus **1**), the exhaust port **137a** can be easily provided above the rotary table **21** near the center of the processing chamber **11**.

(82) The processing region may be a raw material gas adsorption region P1 to which a raw material gas that is to be attached to the substrate is supplied. Accordingly, in the substrate processing apparatus (the deposition apparatus **1**), the raw material gas in the raw material gas adsorption region P1 can be exhausted from the exhaust ports **131a** and **137a** without moving the raw material gas to the lower surface side of the stage **211**.

(83) The processing region may be a reaction gas supply region P2 to which a reaction gas that is to react with the raw material gas attached to the substrate W is supplied. Accordingly, in the substrate processing apparatus (the deposition apparatus **1**), the reaction gas in the reaction gas supply region P2 can be evacuated from the exhaust port **131b** without moving the reaction gas to the lower surface side of the stage **211**.

(84) The processing chamber **11** may include two processing regions separated by the separation regions D. The two processing regions are respectively a raw material gas adsorption region P1 to which a raw material gas that is to attach to the substrate W is supplied, and a reaction gas supply region P2 to which a reaction gas that is to react with the raw material gas attached to the substrate is supplied. Accordingly, in the substrate processing apparatus (the deposition apparatus **1**), the raw material gas and the reaction gas can be evacuated from the exhaust ports **131a**, **131b**, and **137a** while depositing well on the substrate W. As a result, it is possible to prevent mixing of the raw

material gas and the reaction gas on the lower surface side of the stage **211**, thereby preventing the deposition on the stage **211**.

(85) The processing chamber **11** may include two processing regions separated by the separation regions **D**. In one of the two processing regions, a part of the exhaust ports may be provided above the rotary table **21**, and, in another of the two processing regions, an exhaust port may be provided below the rotary table **21**. Even in this case, in the substrate processing apparatus (the deposition apparatus **1**), a plurality of process gases can be prevented from mixing on the lower surface side of the rotary table **21**, thereby avoiding the deposition on the lower surface side of the rotary table **21** and preventing a particle.

(86) An inert gas supplier **27** may be provided that is capable of introducing an inert gas into an under-table space **28** between the rotary table **21** and the processing chamber **11**. The one or more exhaust ports **131a**, **131b**, and **137a** may exhaust the inert gas together with the process gas. Accordingly, in the substrate processing apparatus (the deposition apparatus **1**), the inert gas can flow from the lower surface side of the stage **211** to above the rotary table **21**, thereby more reliably limiting the movement of the process gas.

(87) A substrate processing method according to the second aspect of the present disclosure includes: transferring a substrate **W** to a rotary table **21** provided inside a processing chamber **11**; while rotating the rotary table **21**, relatively rotating a stage **211** that is provided at a position spaced apart from a rotation center of the rotary table **21**, wherein the substrate **W** is placed on the stage; supplying a process gas to the substrate **W** placed on the stage **211** from above the rotary table **21**; supplying a separation gas to separation regions **D**, each of the separation regions being adjacent to a processing region to which the process gas is supplied; and exhausting a gas through one or more exhaust ports **131a**, **131b**, and **137a** that communicate with an inside of the processing chamber **11**. In the exhausting the gas, the gas is exhausted from the one or more exhaust ports **131a**, **131b**, and **137a** that are provided in the processing region between the separation regions **D** and are provided above the rotary table **21**. Even in this case, in the substrate processing method, the process gas supplied into the processing chamber **11** can be prevented from going around to the lower surface side of the stage **211**.

(88) The substrate processing apparatus and the substrate processing method according to the embodiments disclosed herein are exemplary in all respects and are not limiting. Embodiments may be modified and improved in various forms without departing from the scope and spirit of the appended claims. The matters described in the above embodiments may take other configurations within a consistent range, and may be combined within a consistent range.

(89) According to one aspect, the process gas supplied into the processing chamber can be prevented from going around to the lower surface side of the stage.

Claims

1. A substrate processing apparatus comprising: a processing chamber into which a substrate is transferred; a rotary table rotatably provided inside the processing chamber; a stage that is rotatable relative to the rotary table at a position spaced apart from a rotation center of the rotary table, wherein the stage is configured to receive the substrate thereon; a process gas supplier that is provided above the rotary table and that is configured to supply a process gas to the substrate placed on the stage; a separation gas supplier that is configured to supply a separation gas to separation regions, each of the separation regions being adjacent to a processing region to which the process gas is supplied; and a gas exhauster that includes one or more exhaust ports communicating with an inside of the processing chamber and that are configured to exhaust a gas through the one or more exhaust ports, wherein the one or more exhaust ports are provided in the processing region between the separation regions and are provided above the rotary table, wherein at least one of the one or more exhaust ports is provided outside an outer edge of the rotary table,

wherein the gas exhauster includes an outer exhaust structure that is provided outside the outer edge of the rotary table in a radial direction of the rotary table and that is circularly disposed along an inner peripheral surface of the processing chamber, wherein the outer exhaust structure includes a member provided above an upper face of the rotary table in an axial direction of the rotary table, the member having the one or more exhaust ports outside the outer edge of the rotary table in the radial direction of the rotary table, wherein at least one of the one or more exhaust ports that are provided at the member above the upper face of the rotary table open upwardly in the axial direction of the rotary table while the process gas supplier supplies the process gas to the substrate placed on the stage and are provided only in the processing region, wherein the inert gas supplier is configured so that the inert gas flows upwardly from the under-table space to the at least one of the one or more exhaust ports that are provided at the member above the upper face of the rotary table, the rotary table and the outer exhaust structure are arranged so as to form a first gap therebetween, and the rotary table and the stage are arranged so as to form a second gap therebetween, and the inert gas flows upwardly through the first gap and the second gap.

2. The substrate processing apparatus according to claim 1, wherein at least one of the one or more exhaust ports is provided closer to the rotation center of the rotary table than is the stage.
3. The substrate processing apparatus according to claim 2, wherein the processing chamber includes a body having a recess, and a top plate that closes an upper side of the body, and wherein the at least one of the one or more exhaust ports, which is provided closer to the rotation center of the rotary table than is the stage, is provided on the top plate.
4. The substrate processing apparatus according to claim 3, wherein the processing region is a raw material gas adsorption region to which a raw material gas that is to be attached to the substrate is supplied.
5. The substrate processing apparatus according to claim 3, wherein the processing region is a reaction gas supply region to which a reaction gas that is to react with the raw material gas attached to the substrate is supplied.
6. The substrate processing apparatus according to claim 3, wherein the processing chamber includes two processing regions separated by the separation regions, and wherein the two processing regions are respectively a raw material gas adsorption region to which a raw material gas that is to attach to the substrate is supplied, and a reaction gas supply region to which a reaction gas that is to react with the raw material gas attached to the substrate is supplied.
7. The substrate processing apparatus according to claim 3, wherein the processing chamber includes two processing regions separated by the separation regions, and wherein, in one of the two processing regions, the one or more exhaust ports is provided above the rotary table, and, in another of the two processing regions, an exhaust port is provided below the rotary table.
8. The substrate processing apparatus according to claim 4, wherein an inert gas supplier is provided that is capable of introducing an inert gas into an under-table space between the rotary table and the processing chamber, and wherein the one or more exhaust ports exhaust the inert gas together with the process gas.
9. The substrate processing apparatus according to claim 5, wherein an inert gas supplier is provided that is capable of introducing an inert gas into an under-table space between the rotary table and the processing chamber, and wherein the one or more exhaust ports exhaust the inert gas together with the process gas.
10. The substrate processing apparatus according to claim 6, wherein an inert gas supplier is provided that is capable of introducing an inert gas into an under-table space between the rotary table and the processing chamber, and wherein the one or more exhaust ports exhaust the inert gas together with the process gas.
11. The substrate processing apparatus according to claim 1, wherein at least one of the one or more exhaust ports is provided closer to the rotation center of the rotary table than is the stage.
12. The substrate processing apparatus according to claim 11, wherein the processing chamber

includes a body having a recess, and a top plate that closes an upper side of the body, and wherein the at least one of the one or more exhaust ports, which is provided closer to the rotation center of the rotary table than is the stage, is provided on the top plate.

13. The substrate processing apparatus according to claim 1, wherein the processing region is a raw material gas adsorption region to which a raw material gas that is to be attached to the substrate is supplied.

14. The substrate processing apparatus according to claim 1, wherein the processing region is a reaction gas supply region to which a reaction gas that is to react with the raw material gas attached to the substrate is supplied.

15. The substrate processing apparatus according to claim 1, wherein the processing chamber includes two processing regions separated by the separation regions, and wherein the two processing regions are respectively a raw material gas adsorption region to which a raw material gas that is to attach to the substrate is supplied, and a reaction gas supply region to which a reaction gas that is to react with the raw material gas attached to the substrate is supplied.

16. The substrate processing apparatus according to claim 1, wherein the processing chamber includes two processing regions separated by the separation regions, and wherein, in one of the two processing regions, the one or more exhaust ports is provided above the rotary table, and, in another of the two processing regions, an exhaust port is provided below the rotary table.

17. The substrate processing apparatus according to claim 1, wherein an inert gas supplier is provided that is capable of introducing an inert gas into an under-table space between the rotary table and the processing chamber, and wherein the one or more exhaust ports exhaust the inert gas together with the process gas.
